LAB REPORT

CSE312: Database Management System Lab

Submitted To
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Experiment No: 03	Experiment Name: Performing DML Operations on the national Database Using SQL Queries

1. Introduction

This experiment was conducted to practice Data Manipulation Language (DML) operations in a relational database environment using the provided `national.sql` database. The work focused on inserting new data, updating existing data, deleting records based on a condition, and using a subquery with aggregate functions. The experiment helped develop practical skills in executing SQL queries and verifying their results using phpMyAdmin.

2. Objectives

The main objectives of this experiment were:

- To import the `national.sql` database into the local DBMS environment.
- To perform SQL **INSERT**, **UPDATE**, and **DELETE** operations.
- To use a **subquery** for inserting dependent data into a related table.
- To apply an **aggregate function** to delete records based on average order amount.
- To observe and verify query execution results in a real database system.

3. Assigned Tasks

The following tasks were assigned in this lab test:

1. Insert a new region named **South America** into the `region` table.
2. Update the region name **America** to **North America**.
3. Insert a new nation named **Brazil** into the `nation` table under **South America** with a population of **213.5 million**.
4. Delete all the data from the orders table consisting amount between 300 and 780

4. Software and Tools Used

- **Web Server Stack:** XAMPP Control Panel v3.3.0
- **Database Management System:** MariaDB / MySQL
- **Database Interface Tool:** phpMyAdmin
- **Database File:** `national.sql`
- **Text Editor:** Notepad (`232-15-818.txt`)
- **Operating System:** Windows

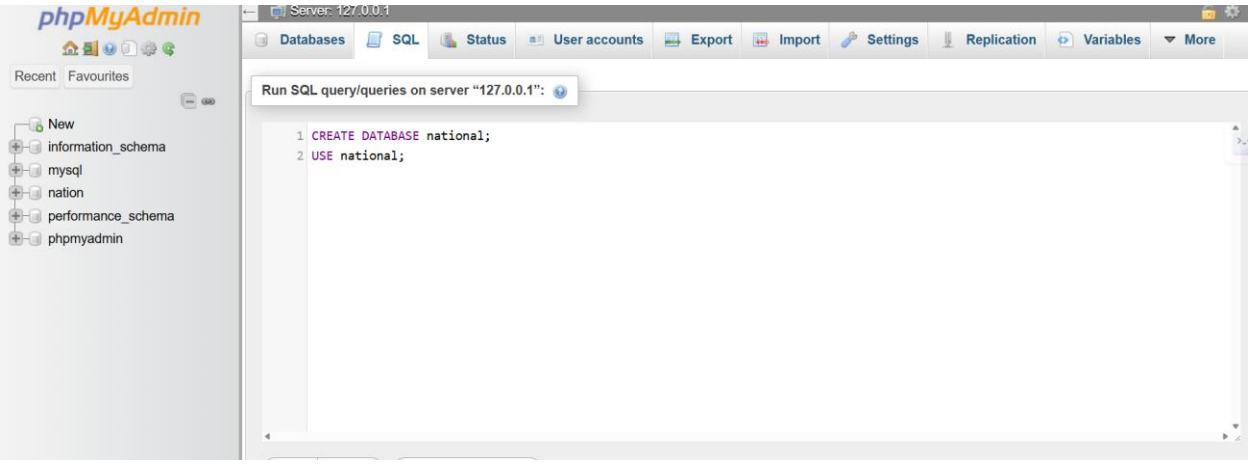
5. Methodology

The experiment was completed in a step-by-step manner:

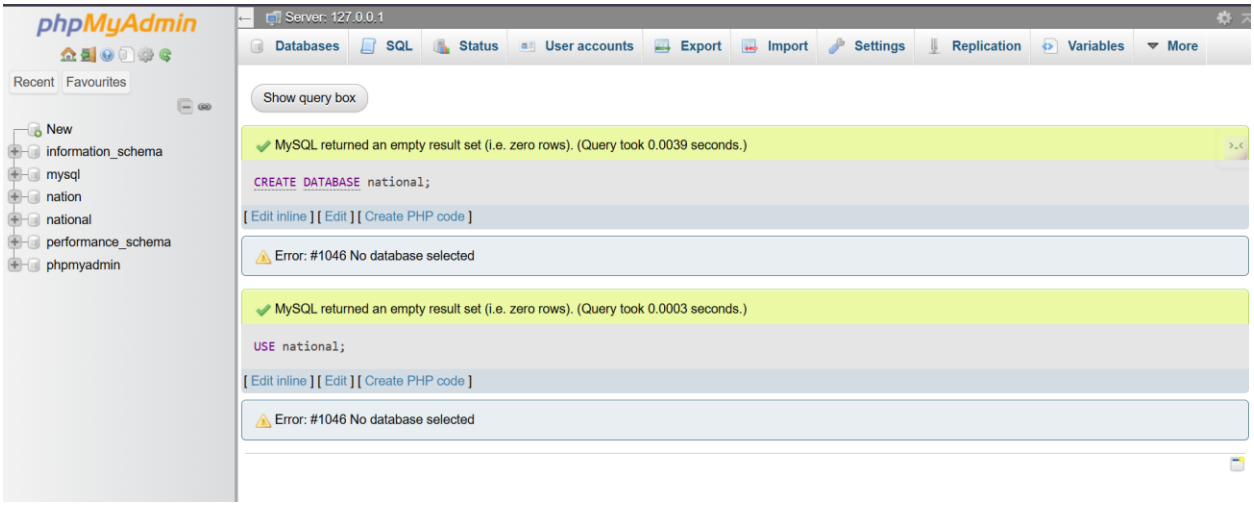
1. The XAMPP server was started and phpMyAdmin was opened.
2. A new database named `national` was created.
3. The provided `national.sql` file was imported successfully into the database.
4. The required tables, especially `region`, `nation`, and `orders`, were examined.
5. All SQL queries were written in a text file named `232-15-818.txt`.
6. Each assigned task was solved one by one using SQL commands.
7. Query execution results were checked carefully after each step.
8. Screenshots of the query, successful execution, and output tables were collected and organized in the report.

6. Experiment Solutions:

Step-01: Create national Database



Step-02: Success of Compilation (national)

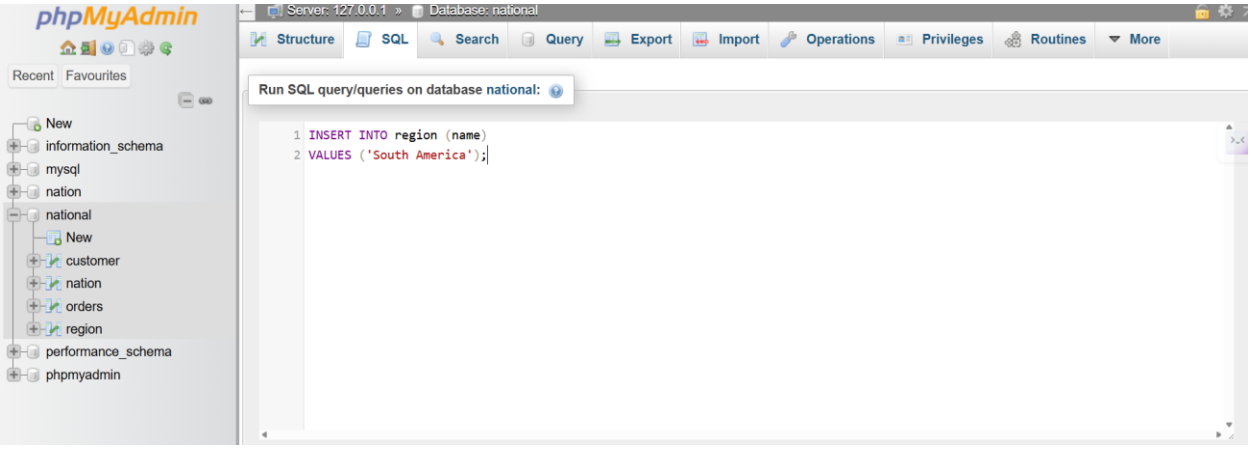


Step-03: Import national.sql file

DML Operations and SQL Queries

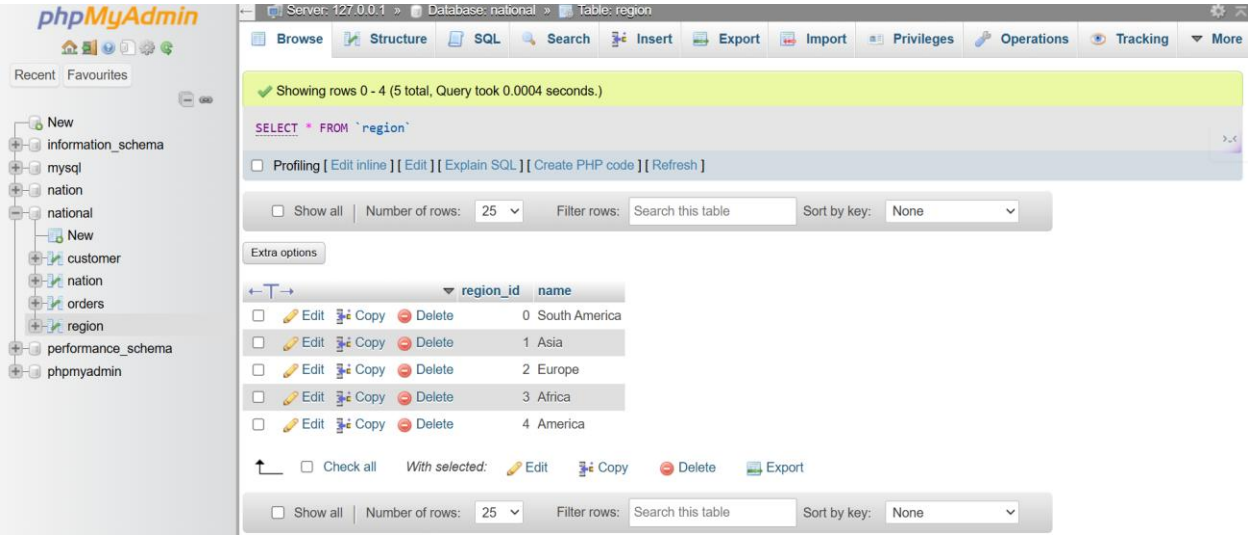
Task 6.1: Insert a new region named "South America" in the region table.

Step-01: SQL Query



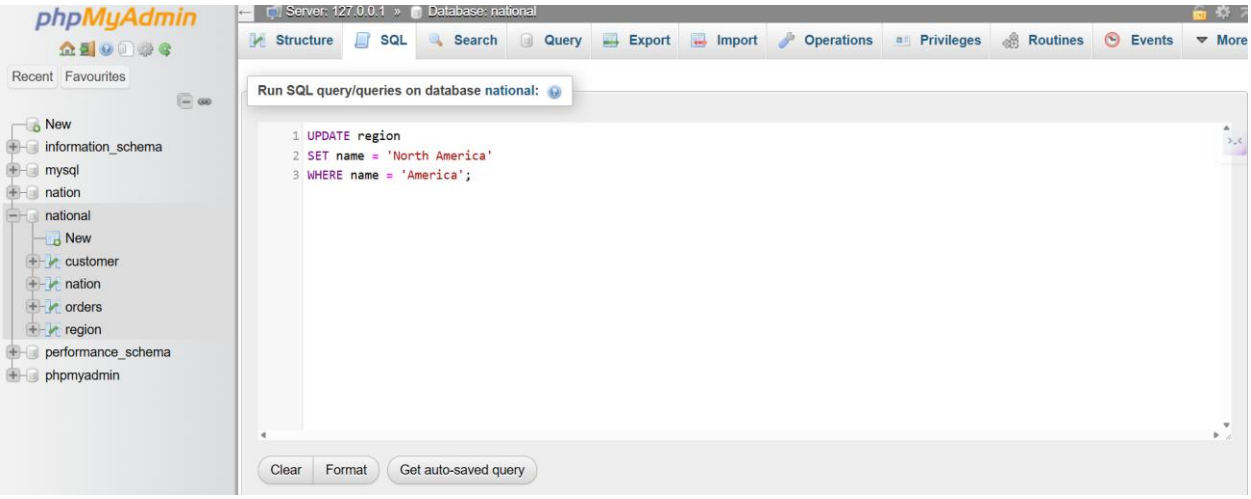
Step-02: Success of Compilation

Step-03: Output (region table after insertion)

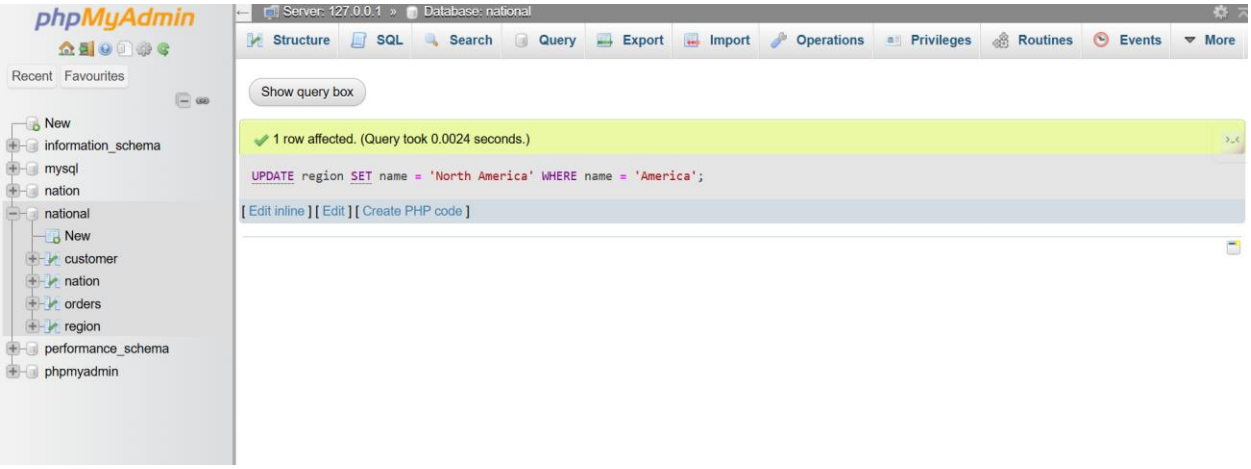


<u>Task 6.2:</u> Update the name of the region named "America" to "North America."

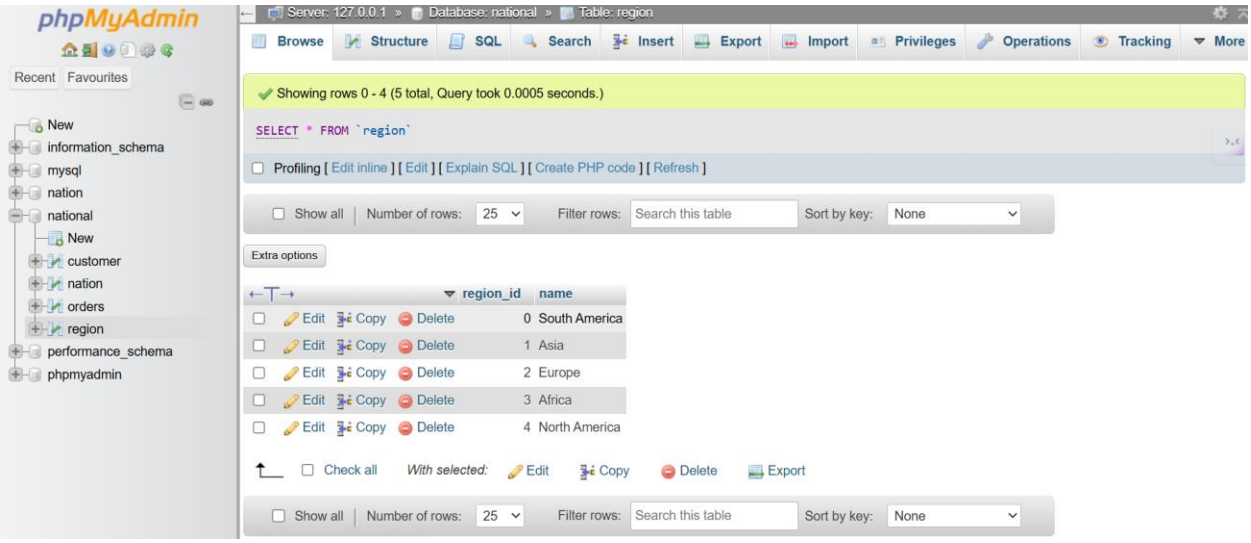
Step-01: SQL Query



Step-02: Success of Compilation

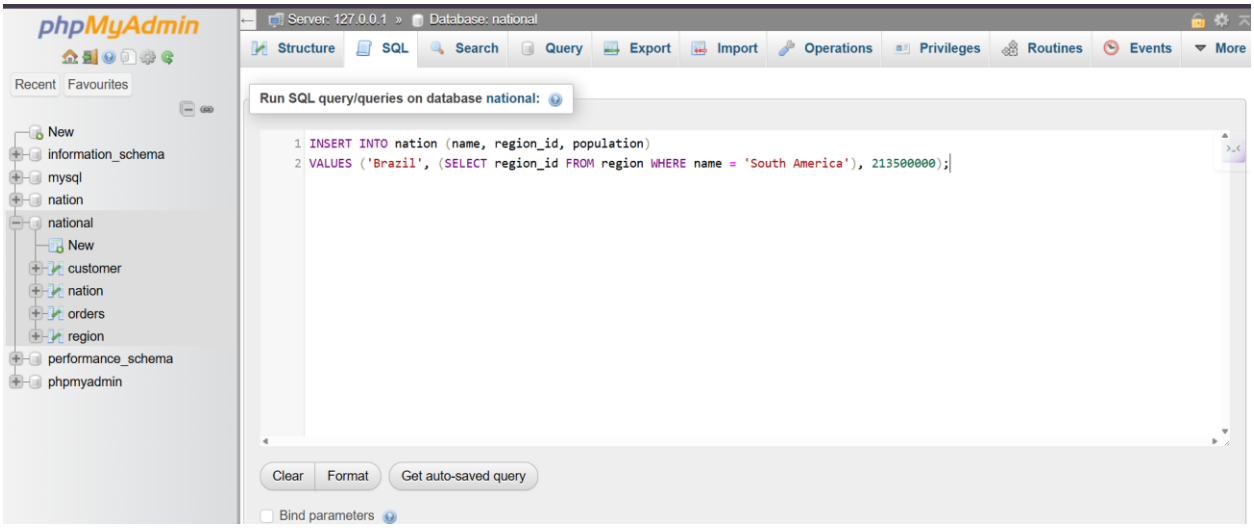


Step-03: Output (region table after update)

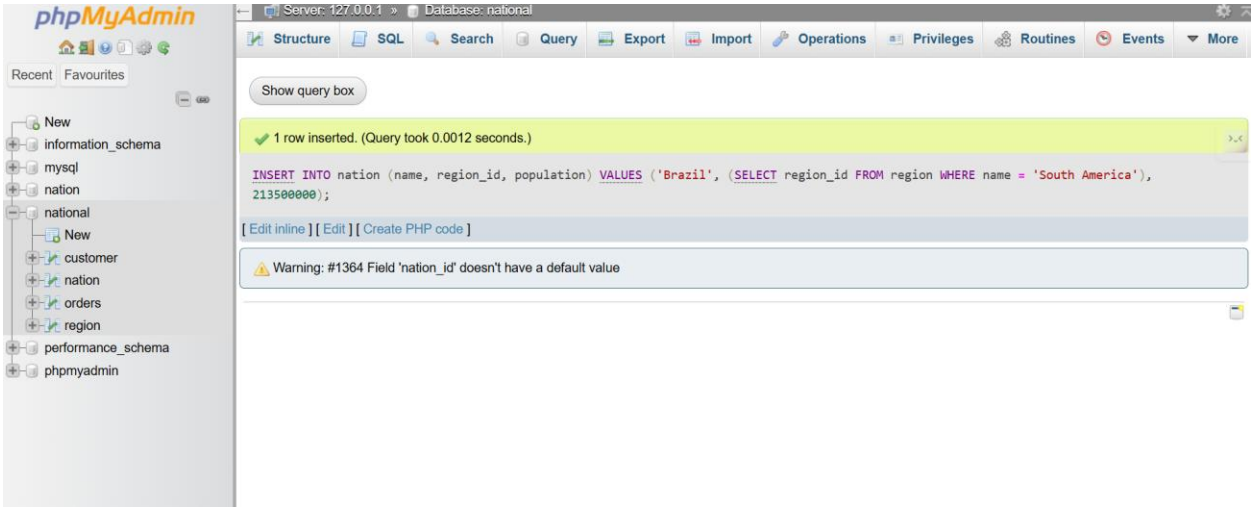


<u>Task 6.3:</u> In the nation table, insert a new entry named "Brazil," which is situated in "South America," with a population of 213.5 million.

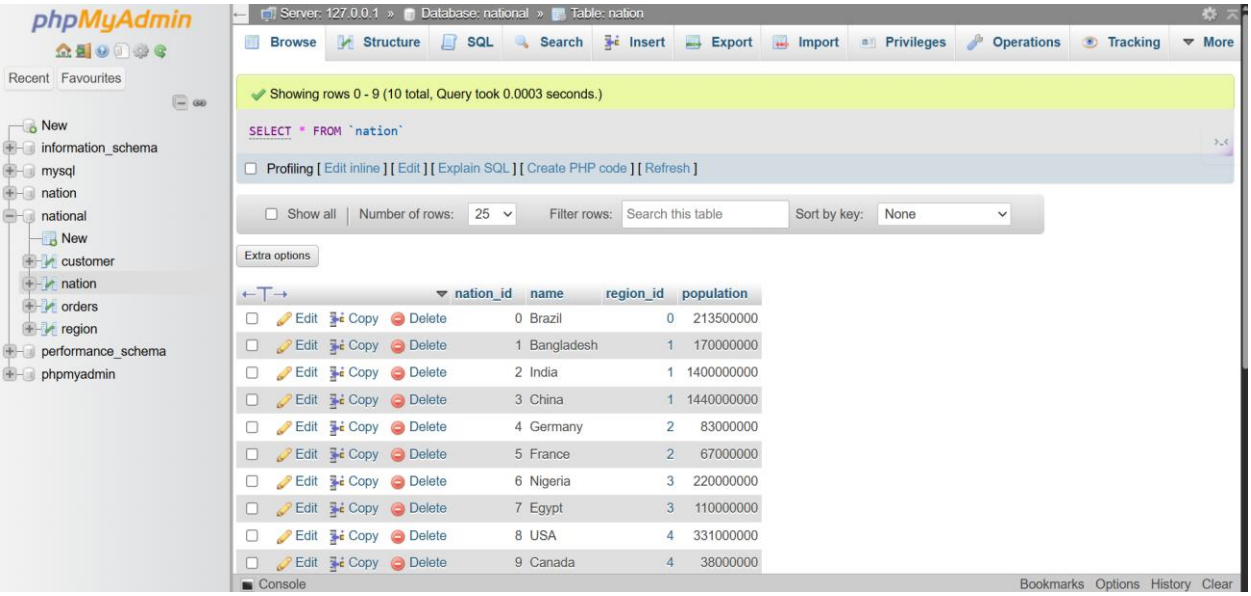
Step-01: SQL Query (Insert into nation)



Step-02: Success of Compilation

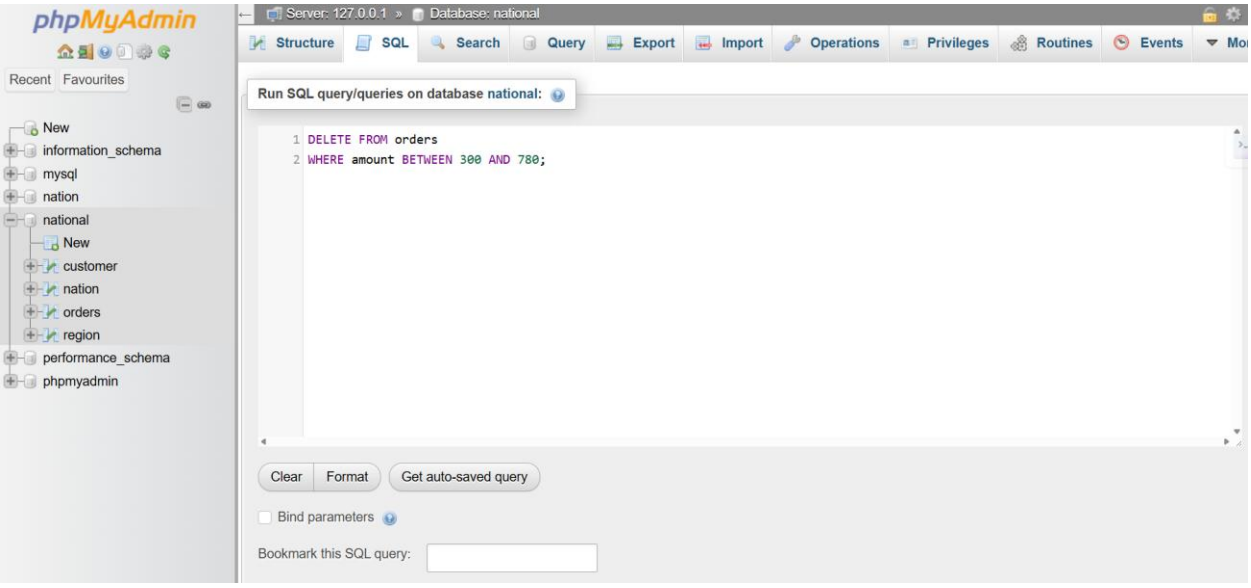


Step-03: Output (nation table after insertion)

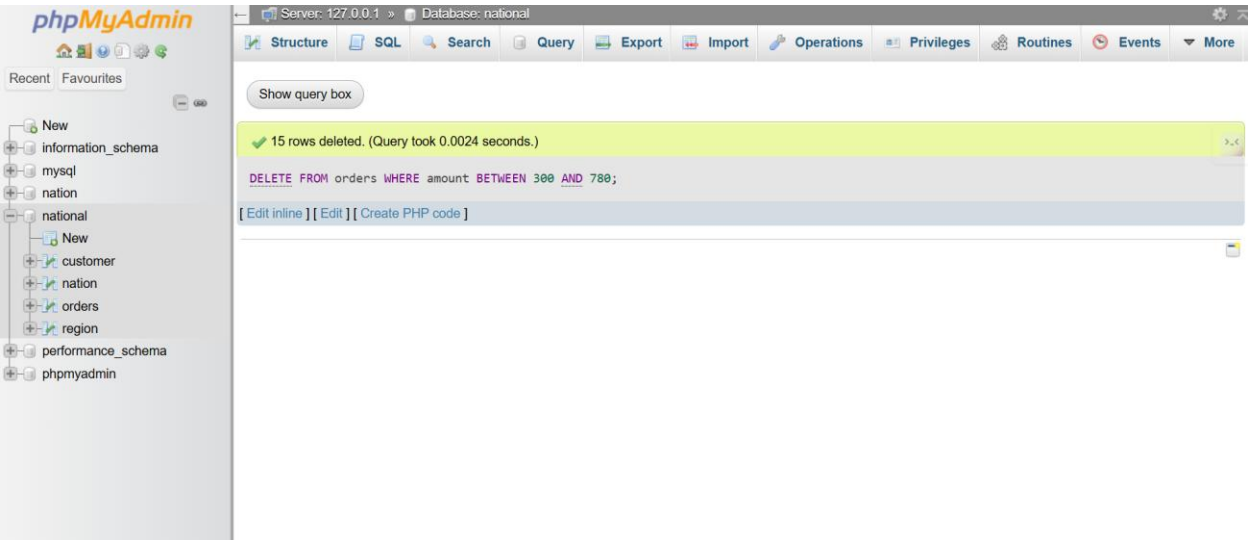


<u>Task 6.4:</u> Delete all the data from the orders table consisting amount between 300 and 780

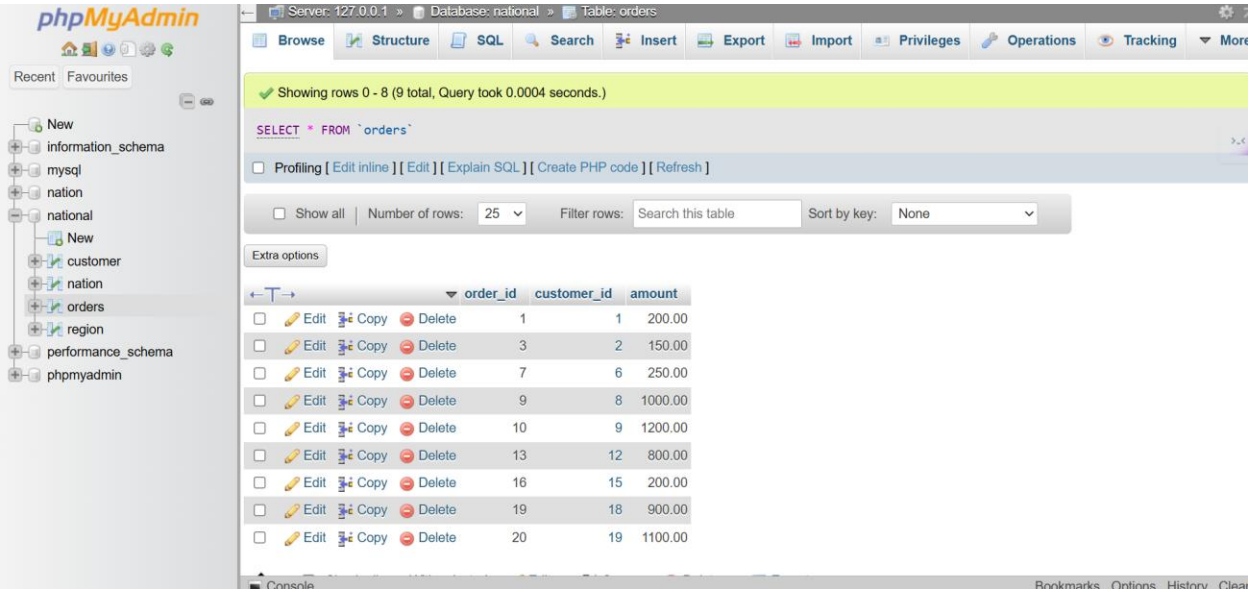
Step-01: SQL Query (Delete from orders)



Step-02: Success of Compilation



Step-03: Output (orders table after deletion)



7. Observation

From this experiment, the following observations were made:

- The `national` database was imported successfully into phpMyAdmin.
- SQL DML commands such as **INSERT**, **UPDATE**, and **DELETE** worked properly.
- The **subquery** was useful for retrieving a dependent value from a related table.
- The **AVG()** aggregate function was successfully used inside a conditional delete statement.
- Practical understanding of relational database operations improved through hands-on execution.
- The experiment demonstrated how real database tables can be manipulated using structured SQL commands.

8. Conclusion

In conclusion, this experiment provided practical knowledge of performing DML operations in a relational database system. By completing insertion, updating, and deletion tasks, the experiment strengthened the understanding of SQL syntax and database manipulation techniques. The use of subqueries and aggregate functions also demonstrated how SQL can solve more dynamic and meaningful problems in database systems. Overall, the lab was successful in achieving its objectives.

9. Appendix

Appendix A: Course Outcome (CO) Attainment

SN	COs	Attainment	Justification
01	CO1	Yes	The lab activities developed practical skills in writing and executing SQL queries for database manipulation and retrieval.
02	CO2	Yes	The experiment used relational concepts, subqueries, and aggregate operations on interconnected tables.
03	CO3	No	Advanced topics such as transaction management, security, and distributed databases were not covered in this experiment.

Appendix B: Addressing Complex Engineering Problems (EP)

SN	Engineering Problem (EP)	Attainment	Related CO	Justification
01	EP1: Depth of Knowledge	Yes	CO1, CO2	The lab required understanding of SQL syntax, query logic, and relational database operations.
02	EP2: Conflicting Requirements	No	CO2	The task was straightforward and did not involve conflicting system constraints.
03	EP4: Familiarity of Issues	Yes	CO1	The experiment addressed standard academic DBMS lab operations.
04	EP7: Interdependence	Yes	CO2	The lab used relations among tables and subqueries dependent on other table values.